Jeonghwan Choi

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Research Interest

My research focuses on 1) *Retrieval-Augmented Generation (RAG)*, designing robust RAG frameworks and developing evaluation methodologies for RAG and LLM reliability in knowledge-intensive tasks, 2) *human-AI alignment* through preference learning, and 3) *LLM-based agent simulation*, where language models are used to simulate complex behaviors and interactions. Beyond these, motivated by my work on agent simulation, I am expanding my research toward *Physical AI*, particularly focusing on autonomous agents for intelligent factory operations and decision-making in fully autonomous manufacturing environments.

Education

- **Korea Advanced Institute of Science and Technology (KAIST)** *Feb 2025 - Present*
  Ph.D. Program, Industrial & Systems Engineering, Advisor: Prof. Hwanjun Song
  (Integrated from M.S.) Daejeon, South Korea
- **Korea Advanced Institute of Science and Technology (KAIST)** *Feb 2024 - Feb 2025*
  M.S., Graduate School of Data Science (GPA: 4.22/4.3), Advisor: Prof. Hwanjun Song Daejeon, South Korea
- **Yonsei University** *Feb 2024*
  B.A. Economics, Statistics Double Major (GPA: 4.21/4.3) Seoul, South Korea

Publications

International Publications

[21] Ready2Blend: From Natural-Language Instructions to Composable Alignment Prompts
Jeesu Jung, Hwan Chang, Juseon Do, **Jeonghwan Choi**, Jinho Choo, Sungwoo Nam, S. K. Hong, Hwanjun Song
Under Review, 2026.

[20] SoCRATES: Towards Reliable Automated Evaluation of Proactive LLM Mediation across Domains and Socio-cognitive Variations
Taewon Yun, Hyeonseong Park, **Jeonghwan Choi**, HaYoon Park, Yeeun Choi, Hwanjun Song
Under Review, 2026.

[19] Toward Agentic Retrieval-Augmented Generation for Enterprise Retail Workflows over Heterogeneous Data Sources
Jeonghwan Choi, Jibin Hwang, HaYoon Park, Hyeonjae Cheon, Hwanjun Song
Under Review, 2026.

[18] Towards Robust Retrieval-Augmented Generation via Reliability-Aware Adaptive Evidence Selection
Minkyu Che, Jihwan Oh, **Jeonghwan Choi**, Jeesu Jung, Hwanjun Song
Findings of the Conference on Empirical Methods in Natural Language Processing (**Findings of EMNLP**), 2026.

[17] What Makes a Sale? Simulating End-to-End Seller–Buyer Retail Dynamics with LLM Agents
Jeonghwan Choi, Jibin Hwang, Gyeonghun Sun, Minjeong Ban, Taewon Yun, Hwanjun Song
Conference on Language Modeling (**COLM**), 2026.

[16] Towards Query-Agnostic RAG Evaluation via Query Coverage and Claim Verifiability
Jeonghwan Choi, Taewon Yun, Minjeong Ban, Gyeonghun Sun, Jae-gil Lee, Hwanjun Song
Conference on Language Modeling (**COLM**), 2026.

[15] RAQE: Reranker-Aligned Query Expansion via Label-Free Group-Relative Policy Optimization
Gyeonghun Sun, **Jeonghwan Choi**, Hwanjun Song
Conference on Language Modeling (**COLM**), 2026.

[14] Distilling Long-CoT Reasoning through Collaborative Step-wise Multi-Teacher Decoding
Taewon Yun, Jisu Shin, **Jeonghwan Choi**, Seunghwan Bang, Hwanjun Song
Findings of the Association for Computational Linguistics (**Findings of ACL**), 2026.

[13] Completing Missing Annotation: Multi-Agent Debate for Accurate and Scalable Relevant Assessment for IR Benchmarks
Jeonghwan Choi*, Minjeong Ban*, Hyangsuk Min*, Nicole Hee-yeon Kim, Minseok Kim, Jae-gil Lee, Hwanjun Song
International Conference on Learning Representations (**ICLR**), 2026.

[12] Aligning Extraction and Generation for Robust Retrieval-Augmented Generation
Hwanjun Song, **Jeonghwan Choi**, Minseok Kim
International Conference on Web Search and Data Mining (**WSDM**), 2026.

[11] BRIDGE: Toward Faithful RAG Benchmarking via Retrieval-Generation Alignment
Jeonghwan Choi*, Minjeong Ban*, Hyangsuk Min*, Hee-yeon Kim, Seunghwan Bang, Hwanjun Song
ACM International Conference on Information and Knowledge Management (**RDGENAI@CIKM**), 2025.

[10] Word2Passage: Word-level Importance Re-weighting for Query Expansion
Jeonghwan Choi, Minjeong Ban, Minseok Kim, Hwanjun Song
Findings of the Association for Computational Linguistics (**Findings of ACL**), 2025.

[9] Learning to Verify Summary Facts with Fine-Grained LLM Feedback
Jihwan Oh, **Jeonghwan Choi**, Nicole Hee-Yeon Kim, Taewon Yun, Hwanjun Song
International Conference on Computational Linguistics (**COLING**), 2025.

Domestic Publications

[8] Toward Low-Cost Query Expansion: Label-Free Reward Design via Reranker Scores
Gyeonghun Sun, **Jeonghwan Choi**, Sundong Kim, Hwanjun Song
Korea Computer Congress (KCC), 2026.

[7] Simulating Seller-Buyer Retail Interactions and Sales Strategies with LLM Agents
Jeonghwan Choi, Jibin Hwang, Gyeonghun Sun, Minjeong Ban, Taewon Yun, Hwanjun Song
Korea Computer Congress (KCC), 2026.

[6] A Query Decompositional Framework for Fine-grained RAG Assessment
Jeonghwan Choi, Minjeong Ban, Taewon Yun, Jihwan Oh, Yuho Lee, Hwanjun Song
KIISE Transactions on Computing Practices, 2026. (*Extended from Korea Software Congress (KSC) 2025,* ***Best Paper***)

[5] A Benchmark Dataset for Retrieval-Augmented Generation Considering Performance Alignment between Retrieval and Generation
Jeonghwan Choi*, Minjeong Ban*, Hyangsuk Min*, Taewon Yun, Jihwan Oh, Hwanjun Song
Korea Computer Congress (KCC), 2025. **(Best Presentation)**

[4] Robust Dataset Condensation via Semi-Supervised Learning
Hee-yeon Kim, **Jeonghwan Choi**, Yuho Lee, Hwanjun Song
Korea Computer Congress (KCC), 2025.

[3] Improving Language Model Quality through LLM-based Fine-Grained Hallucinated Summary Generation
Jihwan Oh, **Jeonghwan Choi**, Nicole Hee-Yeon Kim, Taewon Yun, Hwanjun Song
KIISE Transactions on Computing Practices, 2025.

[2] Training Summary Evaluation Model with Fine-Grained LLM Feedback
Jihwan Oh, **Jeonghwan Choi**, Taewon Yun, Hyangsuk Min, Hwanjun Song
Korea Software Congress (KSC), 2024. **(Best Presentation)**

[1] Improving the Text Summary Quality Through Understanding the Hallucination Level of Summarization Using Large Language Models
Jihwan Oh, **Jeonghwan Choi**, Nicole Hee-Yeon Kim, Hwanjun Song
Korea Computer Congress (KCC), 2024. **(Best Presentation)**

PATENT

- **System and Method for Question Answering over Heterogeneous Data Sources** *Oct 2026*
  Korean Intellectual Property Office (KIPO) Application No: 10-2026-0202245
  - Inventors: **Jeonghwan Choi** (30%), Jibin Hwang (30%), HaYoon Park (10%), Hwanjun Song (30%)

- **An End-to-End Multi-Stage Simulation Framework for Seller-Buyer Retail Dynamics and Sales Strategy Evaluation with LLM Agents** *Sep 2026*
  Korean Intellectual Property Office (KIPO), Provisional Specification Application No: 10-2026-0179065
  - Inventors: **Jeonghwan Choi** (35%), Jibin Hwang (15%), Gyeonghun Sun (5%), Minjeong Ban (5%), Taewon Yun (5%), Hwanjun Song (35%)

- **Reranker-Aligned Query Expansion via Label-Free Group-Relative Policy Optimization** *Jun 2026*
  Korean Intellectual Property Office (KIPO), Provisional Specification Application No: 10-2026-0103506
  - Inventors: Hwanjun Song (45%), Gyeonghun Sun (45%), **Jeonghwan Choi** (10%)

- **Method for Constructing Training Data for a Large Language Model Based on Long-CoT Reasoning Distillation Using Step-Wise Multi-Teacher Collaborative Decoding** *Jun 2026*
  Korean Intellectual Property Office (KIPO) Application No: 10-2026-0100226
  - Inventors: Hwanjun Song (40%), Taewon Yun (40%), **Jeonghwan Choi** (10%), Jisu Shin (10%)

- **A Decompositional Framework for RAG Evaluation through Query Coverage and Claim Verifiability** *May 2026*
  Korean Intellectual Property Office (KIPO), Provisional Specification Application No: 10-2026-0090609
  - Inventors: **Jeonghwan Choi** (35%), Jae-gil Lee (15%), Taewon Yun (5%), Minjeong Ban (5%), Gyeonghun Sun (5%), Hwanjun Song (35%)

- **Method and Apparatus for Document Relevance Assessment Based on Multi-Agent Debate** *Apr 2026*
  Korean Intellectual Property Office (KIPO) Application No: 10-2026-0066617
  - Inventors: **Jeonghwan Choi** (20%), Minjeong Ban (20%), Hyangsuk Min (20%), Hee-yeon Kim (10%), Jae-gil Lee (10%), Hwanjun Song (20%)

- **Method and Apparatus for Generating Answers Using Extended Query-Based Retrieval Utilizing Multi-Level Importance Reweighting** *Sep 2025*
  Korean Intellectual Property Office (KIPO) Application No: 10-2025-0127517
  - Inventors: **Jeonghwan Choi** (40%), Minjeong Ban (20%), Hwanjun Song (40%)

Project Experience

- **National Research Foundation of Korea (NRF)** *Sep 2026 - Aug 2028*
  Principal Investigator - Self-Evolving Physical AI
  - Doctoral Student Research Grant, KRW 50,000,000 over two years

- **Ministry of Science and ICT (IITP)** *Oct 2025 - Present*
  Research Team Member - Multimodal Agentic Platform for Conflict Mediation
  - Developing a multimodal agentic AI for real-time conflict mediation using text, speech, and vision signals

- **GS Retail** *Jan 2026 - June 2026*
  Project Manager - Retrieval-Augmented Generation for Retail
  - Developing a retail-specialized RAG framework with retail environment simulation.

- **National Research Foundation of Korea (NRF)** *Apr 2024 - Present*
  Research Team Member - Language Model Reliability Enhancement System
  - Developing evaluation systems for AI-generated text and reducing hallucinations in LLMs

- **Institute of Information & Communications Technology Planning & Evaluation (IITP)** *Sep 2024 - Present*
  Research Team Member - Korea-Canada Joint Research (KAIST-GIST-UBC)
  - Building value-aligned benchmark datasets for language, math, and medical domains

- **Korea Institute of Science and Technology Information (KISTI)** *Jun 2025 - Dec 2025*
  Project Manager - Scientific Information RAG System
  - Optimized RAG pipeline for scientific domains, maximizing performance while reducing latency

- **Korea Institute of Science and Technology Information (KISTI)** *Jun 2024 - Dec 2024*
  Project Manager - Korean Scientific RAG Framework Development
  - Developed and evaluated RAG framework optimized for Korean scientific domains

Leadership Experience

- **Y-FoRM - Yonsei Forum of Risk Management** *Sep 2022 - Aug 2023*
  President, AI-driven Financial Prediction Research Society
  - Led 30 members and delivered lectures on Statistics, Calculus, Linear Algebra, and Deep Learning

Academic Service

- **2nd Workshop on LLM Agents for Social Simulation (LASS 2026)** *Co-located with ACM CIKM 2026*
  Program Committee Member
- **DocInsights 2026: Workshop on Document Intelligence and Understanding** *Co-located with EMNLP 2026*
  Reviewer

Certifications

- **ADsP (Advanced Data Analytics Semi-Professional)**
  Korea Data Agency March 2022
- **FRM Part 1 (Financial Risk Manager)**
  Global Association of Risk Professionals (GARP) July 2021
- **Investment Asset Management Professional**
  Korea Financial Investment Association June 2020

Honors and Awards

- **Doctoral Student Research Grant (Principal Investigator)**
  National Research Foundation of Korea (NRF) 2026
- **Creative Research Award**
  K-Data Science Conference, Ministry of Science and ICT (MSIT) 2026
- **Best Paper**
  Korea Software Congress (KSC) 2025
- **Best Presentation**
  Korea Computer Congress (KCC) 2025
- **Best Presentation**
  Korea Software Congress (KSC) 2024
- **Best Presentation**
  Korea Computer Congress (KCC) 2024
- **National Scholarship for Humanities Excellence**
  Korea Student Aid Foundation (KOSAF) 2023
- **High Honors (Top 3%)**
  Yonsei University 2023
- **Guwon Foundation Scholarship (Top 1 in Department)**
  Guwon Scholarship Foundation 2023
- **Highest Honors (Top 1%)**
  Yonsei University 2023
- **High Honors (Top 3%)**
  Yonsei University 2022

Skills

- **Languages:** English (Proficient), Korean (Native)
- **Programming:** Python, Java, R

Teaching

- **Dynamic Programming and Reinforcement Learning (IE540)** *KAIST, Spring 2025*
  Teaching Assistant